

Requirement Analysis

Solution Requirements (Functional & Non-functional)

Date	24 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Functional Requirements

The following are the functional requirements of the proposed Smart Agricultural Production Optimization Engine (OptiCrop).

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Crop Recommendation	Accept soil nutrient values (N, P, K), temperature, humidity, pH, and rainfall from the user.
		Validate all user inputs before prediction.
		Predict the most suitable crop using the trained machine learning model.
FR-2	Crop Suitability Evaluation	Allow users to select a crop and evaluate whether it is suitable under the given environmental conditions.
		Display suitability status and recommendation.

FR-3	Agricultural Data Analysis	Analyze the agricultural dataset to identify crop patterns and relationships.
		Compare different machine learning algorithms based on prediction accuracy.
FR-4	Machine Learning Model Management	Train multiple machine learning algorithms.
		Select the best-performing model.
		Save the trained model as a Pickle (.pkl) file for deployment.
FR-5	Web Application Interface	Provide a simple web interface for user interaction.
		Display the predicted crop recommendation clearly.
		Handle invalid or incomplete user inputs gracefully.

Non-Functional Requirements

The following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, user-friendly, and easy-to-understand interface for farmers and researchers.
NFR-2	Security	User inputs should be validated to prevent invalid data and ensure reliable predictions.
NFR-3	Reliability	The system should consistently provide accurate crop recommendations based on the trained machine learning model.
NFR-4	Performance	The application should generate predictions within a few seconds by loading a pre-trained model instead of retraining it.

NFR-5	Availability	The application should remain accessible whenever the Flask server is running.
NFR-6	Scalability	The modular architecture should allow future integration of additional datasets, crop varieties, or machine learning algorithms without major redesign.

